

EXPERIMENT 17

Network Commands Execution (ping, traceroute, ifconfig, netstat)

Aim

To execute basic network diagnostic commands (ping, traceroute, ifconfig/ip, netstat/ss) on Linux and observe their output to understand network connectivity, routing path, local IP configuration and active connections.

Procedure

1. ping – Test Connectivity

```
ping -c 4 google.com
```

2. traceroute – Find Path / Hops

```
traceroute google.com
```

3. ifconfig / ip – View IP Configuration

```
ifconfig  
ip addr
```

4. netstat / ss – View Active Connections

```
netstat -tuln  
ss -tuln
```

Output

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ipconfig  
Command 'ipconfig' not found, did you mean:  
  command 'ifconfig' from deb net-tools  
  command 'iwconfig' from deb wireless-tools  
  command 'hipconfig' from deb hipcc  
  command 'iconfig' from deb ipmiutil  
Try: sudo apt install <deb name>  
  
(kali@kali)-[~]  
$ ping -c 4 google.com  
PING google.com (172.217.24.14) 56(84) bytes of data.  
64 bytes from lcmaaa-an-in-f14.1e100.net (172.217.24.14): icmp_seq=1 ttl=64 time=7.76 ms  
64 bytes from lcmaaa-an-in-f14.1e100.net (172.217.24.14): icmp_seq=2 ttl=64 time=44.4 ms  
64 bytes from lcmaaa-an-in-f14.1e100.net (172.217.24.14): icmp_seq=3 ttl=64 time=10.6 ms  
64 bytes from lcmaaa-an-in-f14.1e100.net (172.217.24.14): icmp_seq=4 ttl=64 time=16.1 ms  
  
— google.com ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 7.763/19.721/44.434/14.578 ms  
  
(kali@kali)-[~]  
$ tracert google.com  
Command 'tracert' not found, did you mean:  
  command 'tracert6' from deb ndisc6  
Try: sudo apt install <deb name>  
  
(kali@kali)-[~]  
$ traceroute google.com  
traceroute to google.com (142.251.221.174), 30 hops max, 60 byte packets  
 1  10.0.2.2 (10.0.2.2)  2.223 ms  2.039 ms  1.961 ms  
 2  * * *  
 3  * * *  
 4  * * *  
 5  * * *  
 6  * * *  
 7  * * *  
 8  * * *  
 9  * * *  
10  * * *  
11  * * *  
12  * * *  
13  * * *  
14  * * *  
15  * * *  
16  * * *  
17  * * *  
18  * * *  
19  * * *  
20  * * *
```

```
kali@kali: ~
Session Actions Edit View Help
└─$
(kali@kali)-[~]
└─$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps        ESTABLISHED
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags               Type               State         I-Node    Path
unix   3      [ ]                 STREAM            CONNECTED      13715      @/tmp/.X11-unix/X0
unix   3      [ ]                 STREAM            CONNECTED      11476      /run/systemd/journal/stdout
unix   3      [ ]                 STREAM            CONNECTED      10553      /run/systemd/journal/stdout
unix   3      [ ]                 STREAM            CONNECTED      14893      /run/user/1000/pipewire-0
unix   3      [ ]                 STREAM            CONNECTED      15659
unix   3      [ ]                 STREAM            CONNECTED      13741      @/tmp/.X11-unix/X0
unix   3      [ ]                 STREAM            CONNECTED      11534      /run/user/1000/bus
unix   3      [ ]                 STREAM            CONNECTED      10869      /run/user/1000/bus
unix   3      [ ]                 STREAM            CONNECTED      14345      /run/user/1000/pulse/native
unix   3      [ ]                 STREAM            CONNECTED      11670
unix   3      [ ]                 STREAM            CONNECTED      9366
unix   3      [ ]                 STREAM            CONNECTED      14891      /run/user/1000/bus
unix   3      [ ]                 STREAM            CONNECTED      11198
unix   3      [ ]                 STREAM            CONNECTED      15884
unix   3      [ ]                 STREAM            CONNECTED      12446
unix   3      [ ]                 STREAM            CONNECTED      11486      /run/systemd/journal/stdout
unix   3      [ ]                 STREAM            CONNECTED      10843
unix   3      [ ]                 STREAM            CONNECTED      8698
unix   3      [ ]                 STREAM            CONNECTED      14269
unix   3      [ ]                 STREAM            CONNECTED      14815
unix   3      [ ]                 STREAM            CONNECTED      12578      /run/user/1000/at-spi/bus_0
unix   3      [ ]                 STREAM            CONNECTED      11047      /run/dbus/system_bus_socket
unix   2      [ ACC ]             STREAM            LISTENING       10988      /home/kali/.ssh/agent/s.RDJpj98jBr.agent.07FB0QU34Y
unix   3      [ ]                 STREAM            CONNECTED      10855      /run/systemd/journal/stdout
unix   3      [ ]                 STREAM            CONNECTED      14571
unix   3      [ ]                 STREAM            CONNECTED      8159      /run/user/1000/bus
unix   3      [ ]                 STREAM            CONNECTED      6718
unix   3      [ ]                 STREAM            CONNECTED      13733      @/tmp/.X11-unix/X0
unix   3      [ ]                 STREAM            CONNECTED      13343
unix   3      [ ]                 STREAM            CONNECTED      14202
unix   3      [ ]                 STREAM            CONNECTED      16790      @/tmp/.X11-unix/X0
unix   3      [ ]                 STREAM            CONNECTED      11665      @/tmp/.X11-unix/X0
unix   3      [ ]                 STREAM            CONNECTED      11414      /run/systemd/journal/stdout
unix   3      [ ]                 STREAM            CONNECTED      9149      /run/systemd/journal/stdout
unix   3      [ ]                 DGRAM            CONNECTED      4708
unix   3      [ ]                 STREAM            CONNECTED      15710      /run/user/1000/bus
unix   3      [ ]                 STREAM            CONNECTED      10198
unix   3      [ ]                 STREAM            CONNECTED      8731
unix   3      [ ]                 STREAM            CONNECTED      6094
unix   2      [ ACC ]             STREAM            LISTENING       6130      /tmp/.iprt-localipc-DRMIpcServer
unix   3      [ ]                 STREAM            CONNECTED      8259      /run/systemd/journal/stdout
```

```

kali@kali: ~
Session Actions Edit View Help
unix 3 [ ] STREAM CONNECTED 9974
unix 3 [ ] STREAM CONNECTED 14243
unix 3 [ ] STREAM CONNECTED 14916 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 8184
unix 3 [ ] STREAM CONNECTED 12081 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 14696 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 14669
unix 3 [ ] STREAM CONNECTED 9350 /run/systemd/journal/stdout
unix 3 [ ] STREAM CONNECTED 12103 /tmp/.ICE-unix/1025
unix 3 [ ] STREAM CONNECTED 13333 /run/user/1000/at-spi/bus_0
unix 3 [ ] STREAM CONNECTED 6722 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 13337
unix 3 [ ] STREAM CONNECTED 14583
unix 3 [ ] STREAM CONNECTED 8142 /run/user/1000/bus
unix 3 [ ] STREAM CONNECTED 11273
unix 3 [ ] STREAM CONNECTED 11197 /run/user/1000/at-spi/bus_0
unix 3 [ ] STREAM CONNECTED 13759 /run/user/1000/bus
unix 3 [ ] STREAM CONNECTED 7360 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 6719
unix 3 [ ] STREAM CONNECTED 9882 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 16339
unix 3 [ ] STREAM CONNECTED 9925
unix 3 [ ] STREAM CONNECTED 6610
unix 3 [ ] STREAM CONNECTED 12030
unix 3 [ ] STREAM CONNECTED 14609 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 13340
unix 3 [ ] STREAM CONNECTED 11635 @/tmp/.X11-unix/X0
unix 3 [ ] STREAM CONNECTED 11067
unix 3 [ ] STREAM CONNECTED 9874
unix 3 [ ] STREAM CONNECTED 16406
unix 3 [ ] STREAM CONNECTED 12100 /run/systemd/journal/stdout
unix 3 [ ] STREAM CONNECTED 11305
unix 3 [ ] STREAM CONNECTED 9914
unix 3 [ ] STREAM CONNECTED 13823
unix 3 [ ] DGRAM CONNECTED 4706
unix 3 [ ] STREAM CONNECTED 18542
unix 3 [ ] STREAM CONNECTED 13514
unix 3 [ ] STREAM CONNECTED 10941
unix 3 [ ] DGRAM CONNECTED 7979
unix 2 [ ACC ] STREAM LISTENING 12291 @/tmp/.ICE-unix/1025
unix 2 [ ] DGRAM 4867 @14460866311037769630
unix 2 [ ACC ] STREAM LISTENING 9363 @/tmp/.X11-unix/X0
unix 3 [ ] STREAM CONNECTED 7981 @8a964461f5ccb092/bus/systemd/bus-system
unix 2 [ ] STREAM 12106 @printer-applet-lock-user-kali
unix 3 [ ] STREAM CONNECTED 11277 @9f35370754fe5b32/bus/systemd/bus-api-user
unix 3 [ ] STREAM CONNECTED 6729 @b42e9494fe828483/bus/systemd/bus-api-system
unix 3 [ ] STREAM CONNECTED 8210 @735b60ab78cb4cb2/bus/systemd-logind/system

(kali@kali)-[~]
$ hostname
kali

```